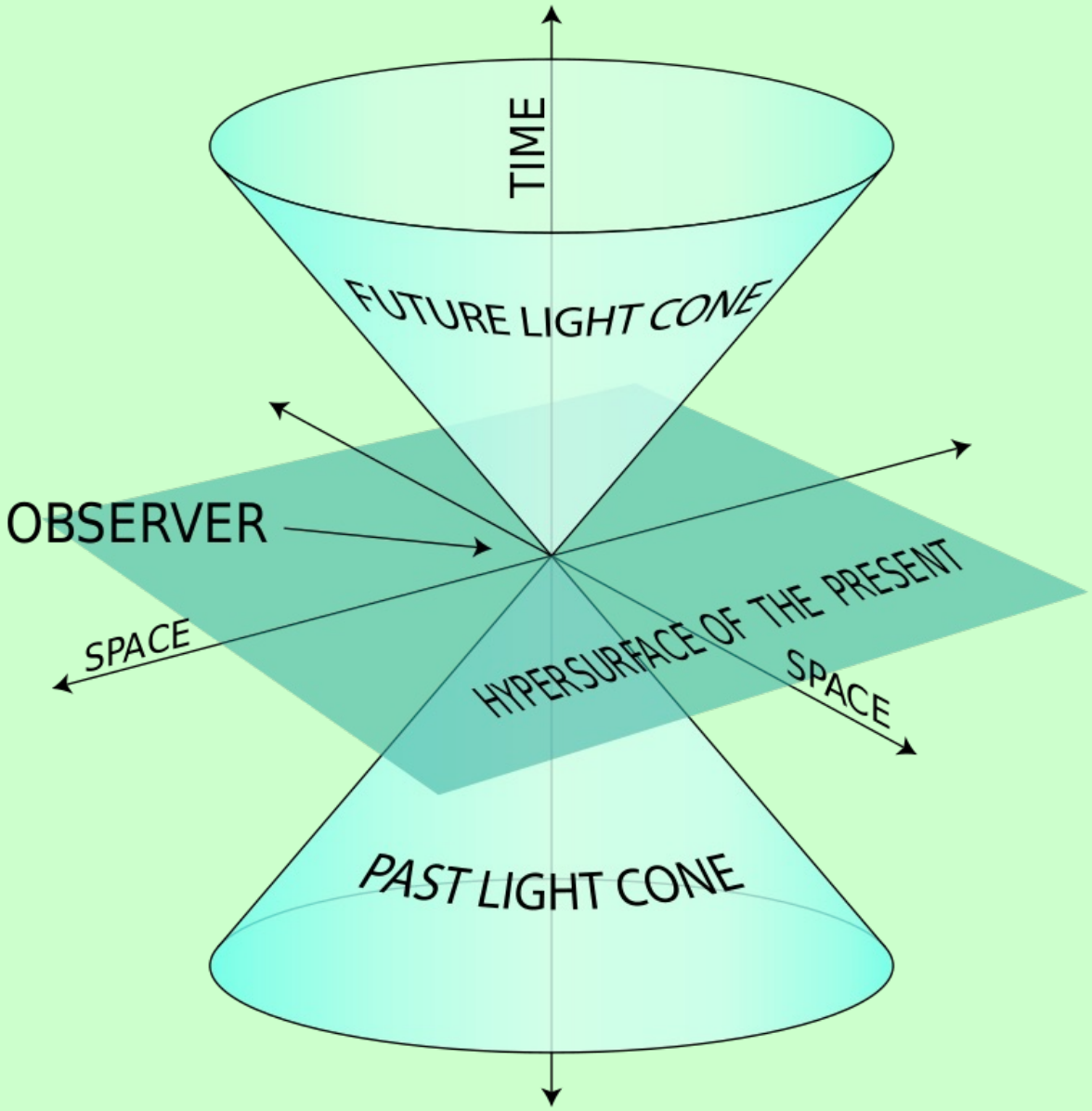
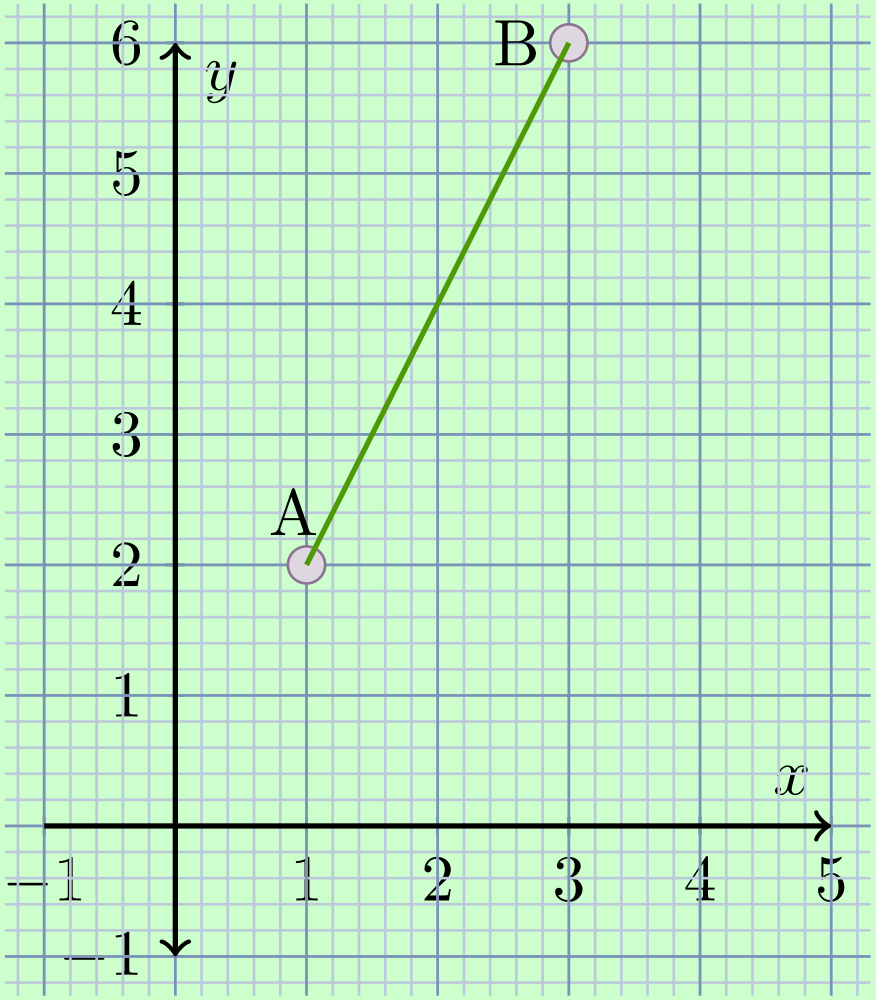


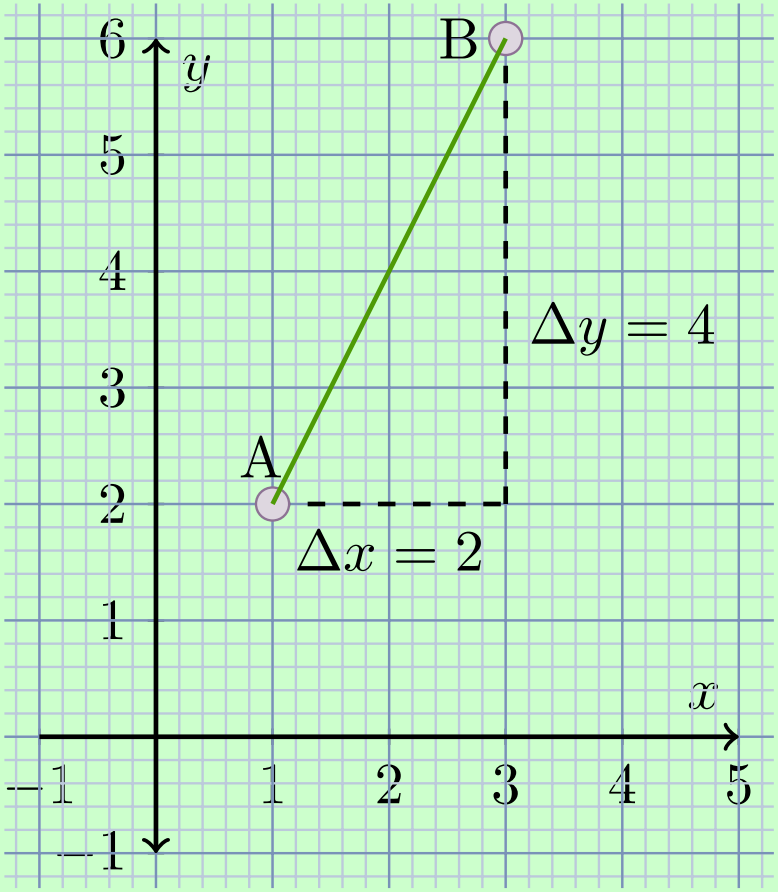
Spacetime Diagrams

Quick Quiz

How far from A to B?



Use the Pythagorean theorem!!



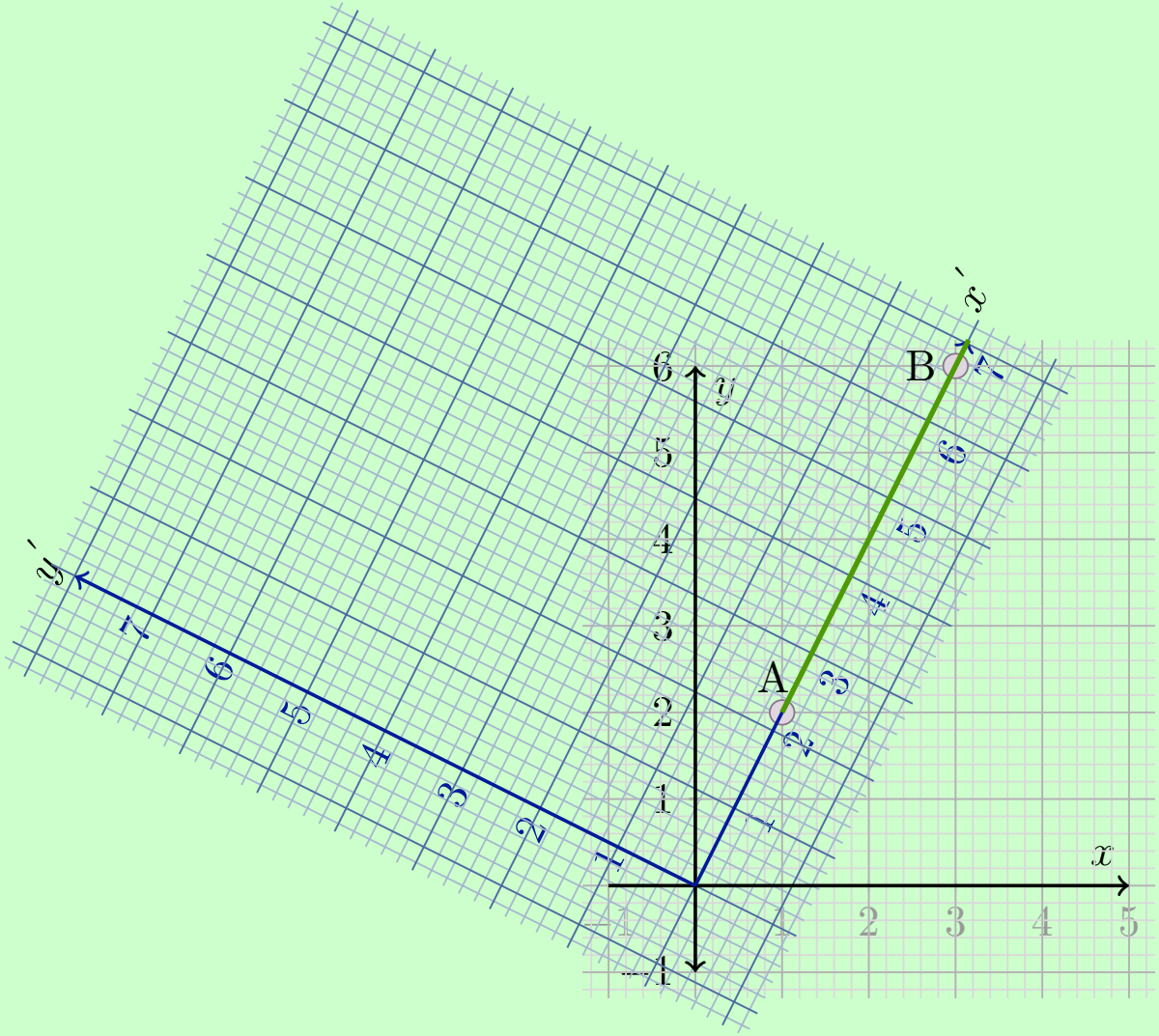
$$\Delta d^2 = \Delta x^2 + \Delta y^2$$

$$\Delta d = \sqrt{(2)^2 + (4)^2}$$

$$\Delta d = \sqrt{20} = 4.47$$

Or don't.
Just rotate A and B are now parallel to a a new x′ axis.

Now just read it off:

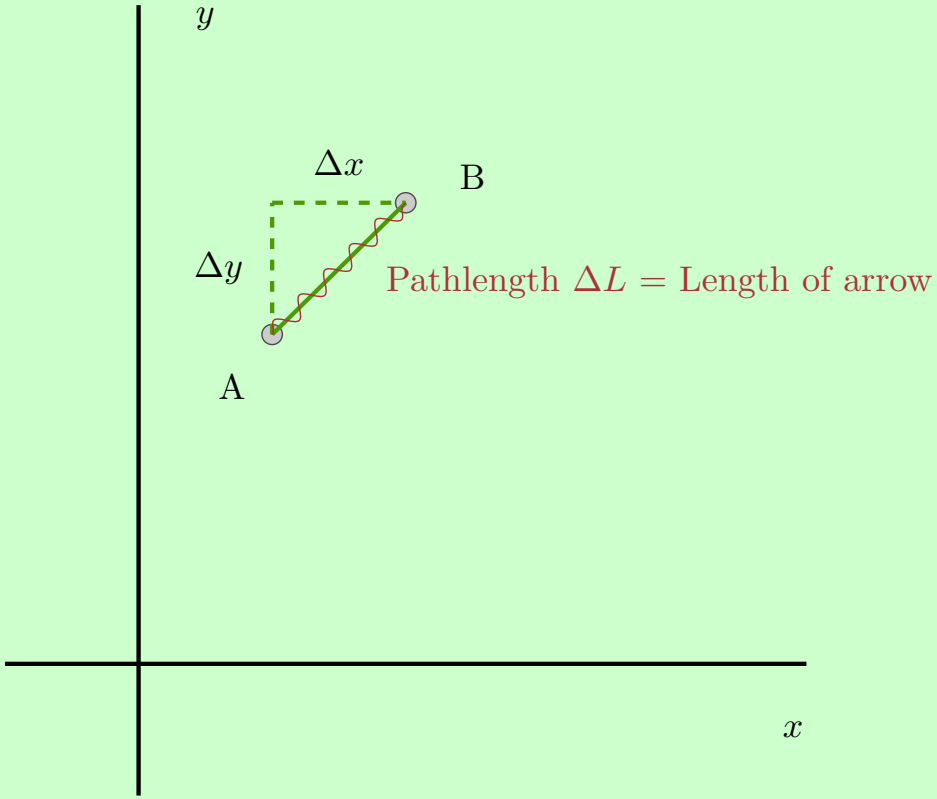


$$\Delta d^2 = \Delta x'^2 + \Delta y'^2$$
$$\Delta d = \sqrt{(4.47)^2 + (0)^2}$$
$$\Delta d = 4.47$$

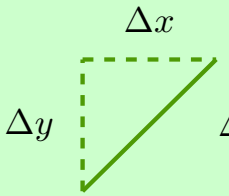
In normal geometry distance is invariant:

$$\Delta d^2 = \Delta x^2 + \Delta y^2$$
$$= \Delta x'^2 + \Delta y'^2$$

Geometric Analogy: 2D space



Cartesian



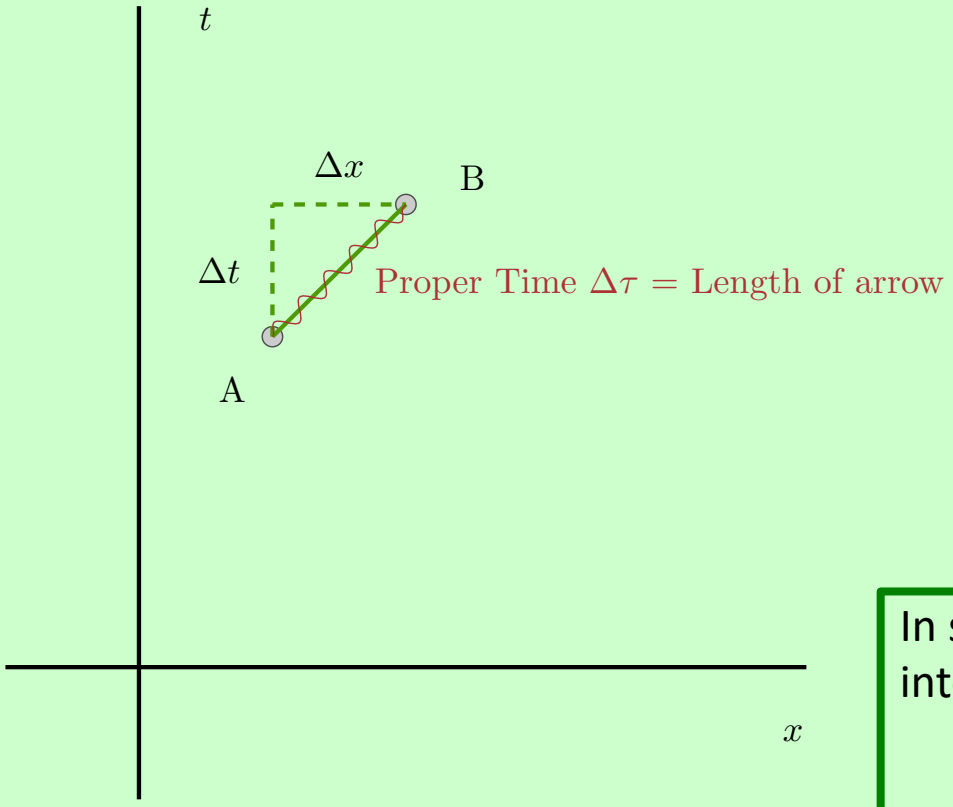
$$\Delta d^2 = \Delta x^2 + \Delta y^2$$

In normal geometry distance is invariant:

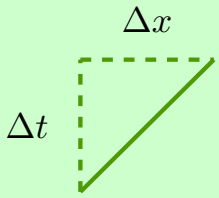
$$\Delta d^2 = \Delta x^2 + \Delta y^2$$
$$= \Delta x'^2 + \Delta y'^2$$

Some quantities are relative (d, v), some are invariant (m, F, spacetime intervals)

Geometric Analogy: 1+1 Spacetime



Spacetime



$$\Delta s^2 = \Delta x^2 - \Delta t^2$$

In spacetime, the spacetime interval is invariant:

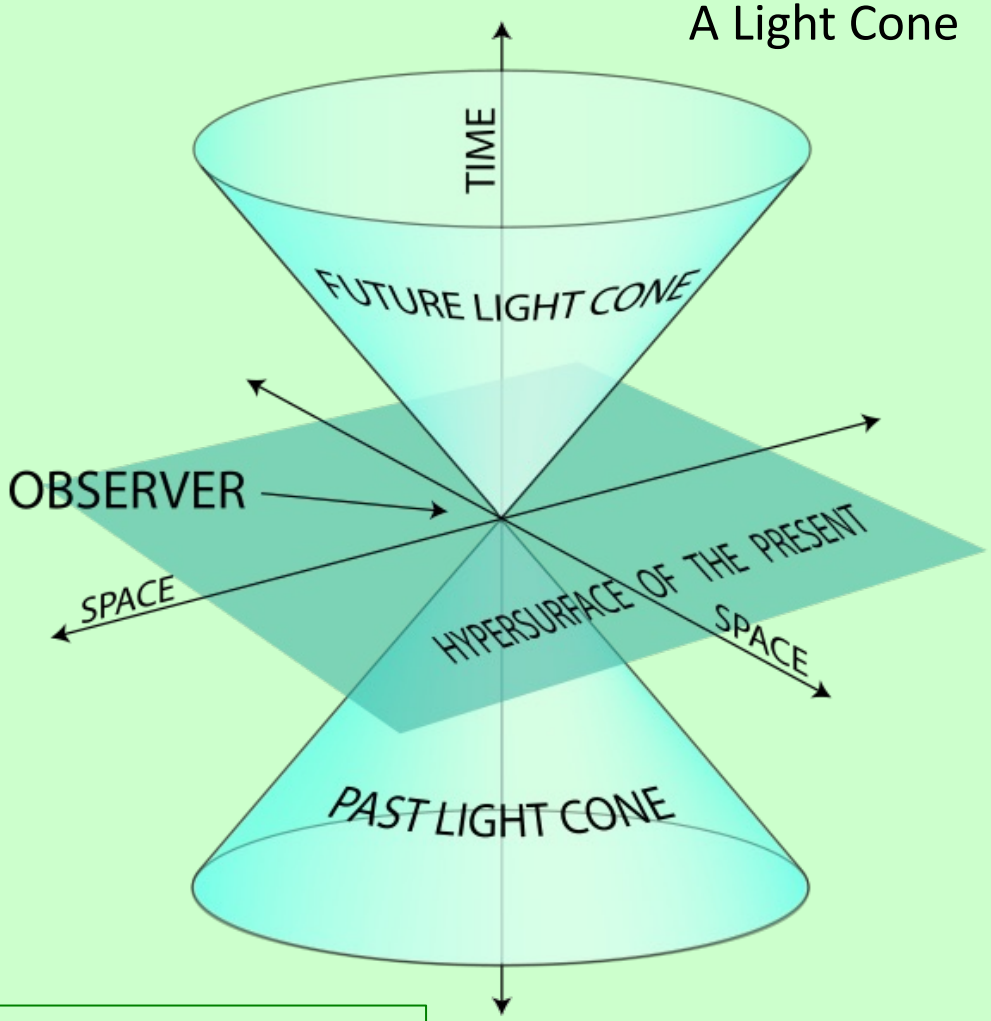
$$\Delta s^2 = \Delta x^2 - \Delta t^2 = \Delta x^{2'} - \Delta t^{'2}$$

A and B are called “events”

Spacetime

1. Light moves at 45^0, on the boundary of the light-cone
2. Past light cone = events that can influence the observer
3. Future light cone are event the observer can influence

A Light Cone



The spacetime Interval is invarient:

$$\Delta s^2 = \Delta x^2 - \Delta t^2$$

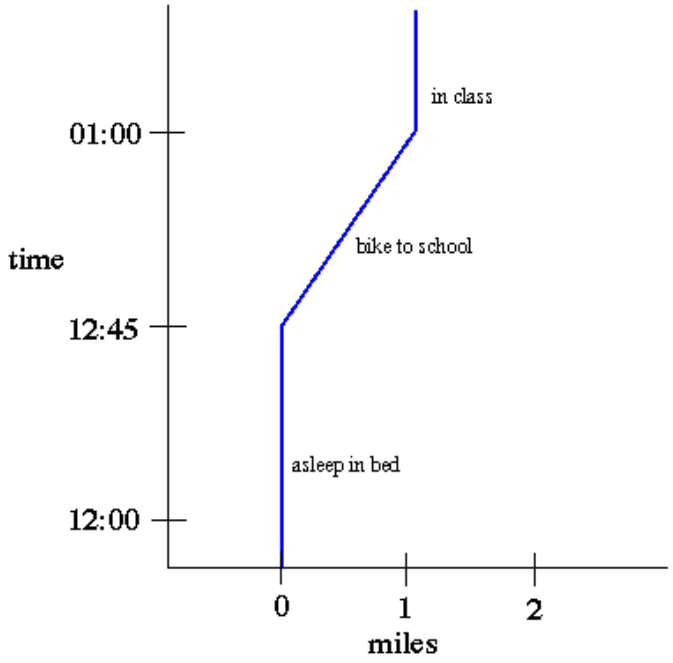
Two observers in won't agree on how far away something is
Two observers in won't agree on when something happened
They do agree on the spacetime interval

Ex. Draw worldlines of:

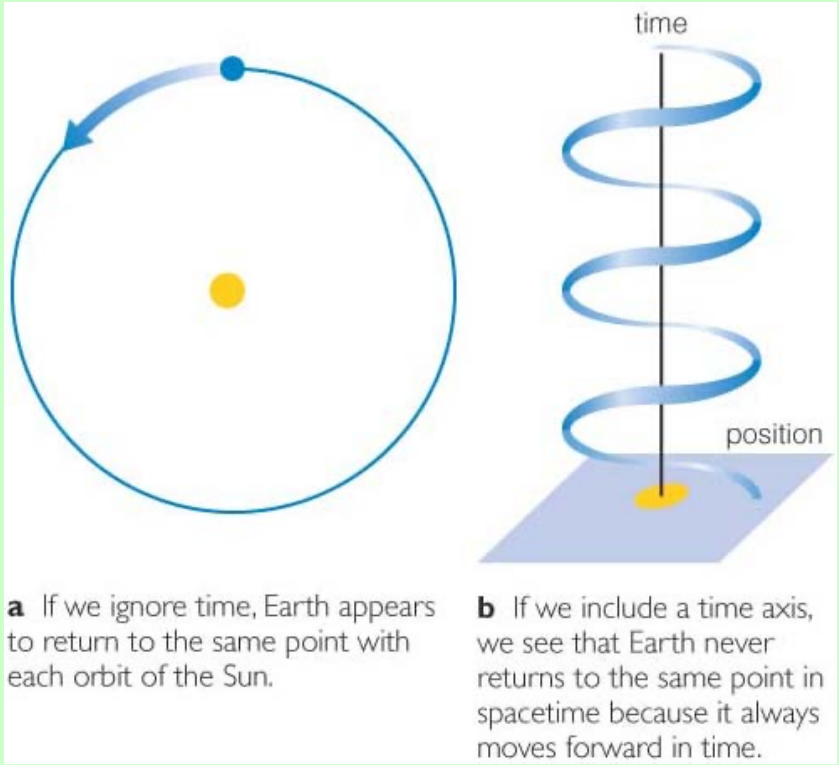
1) your day, and estimate your Δs
2) the motion of the earth around the sun

World Lines

A world line is the plot of an event or object in spacetime. Typically only one of the 3 spatial coordinates are shown in order to make a flat plot



My Daily worldline

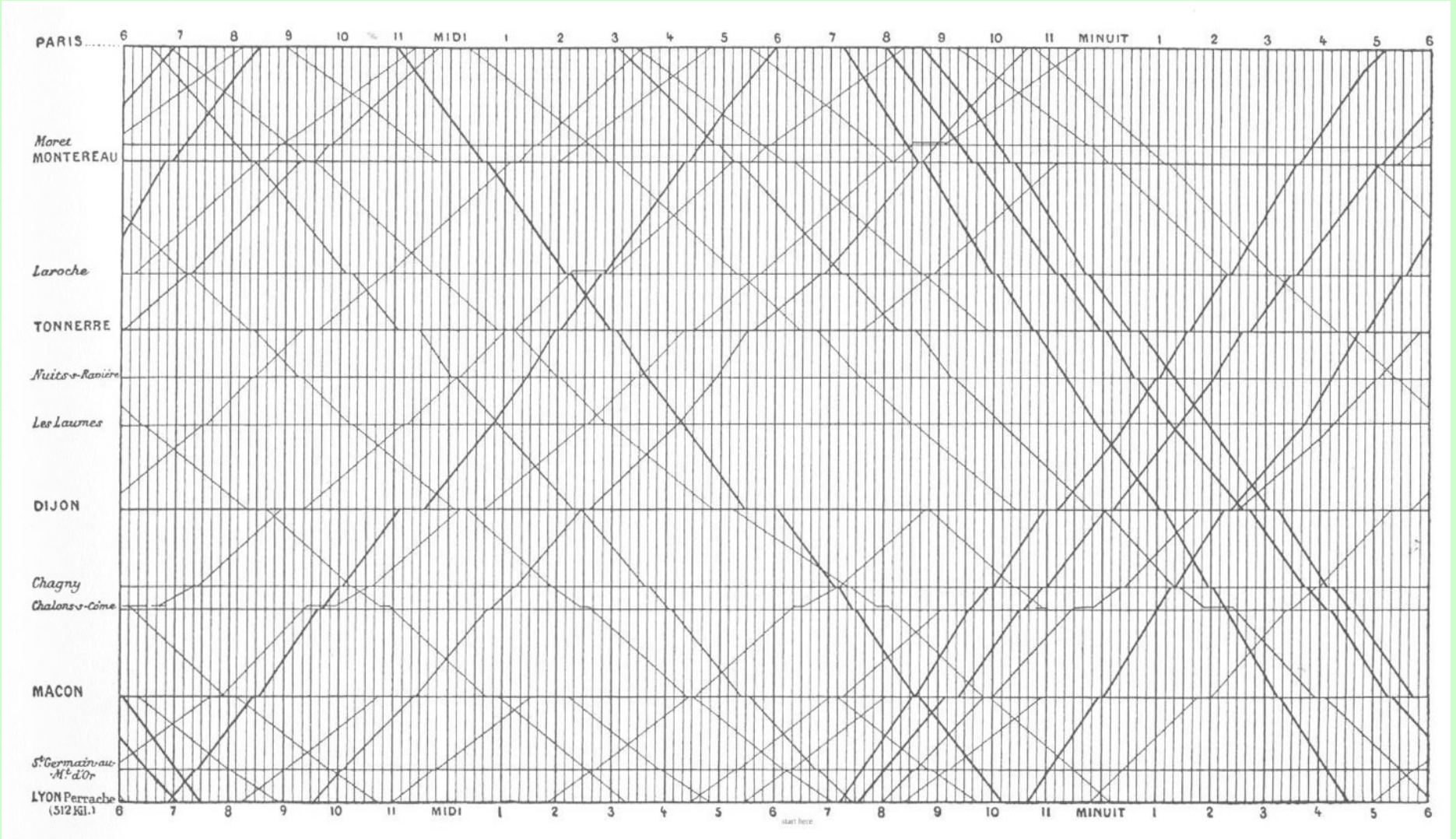


a If we ignore time, Earth appears to return to the same point with each orbit of the Sun.

b If we include a time axis, we see that Earth never returns to the same point in spacetime because it always moves forward in time.

The Earth's Worldline

French Train Worldlines



A. What happens when paths cross?

B. Which trains are the Fastest?

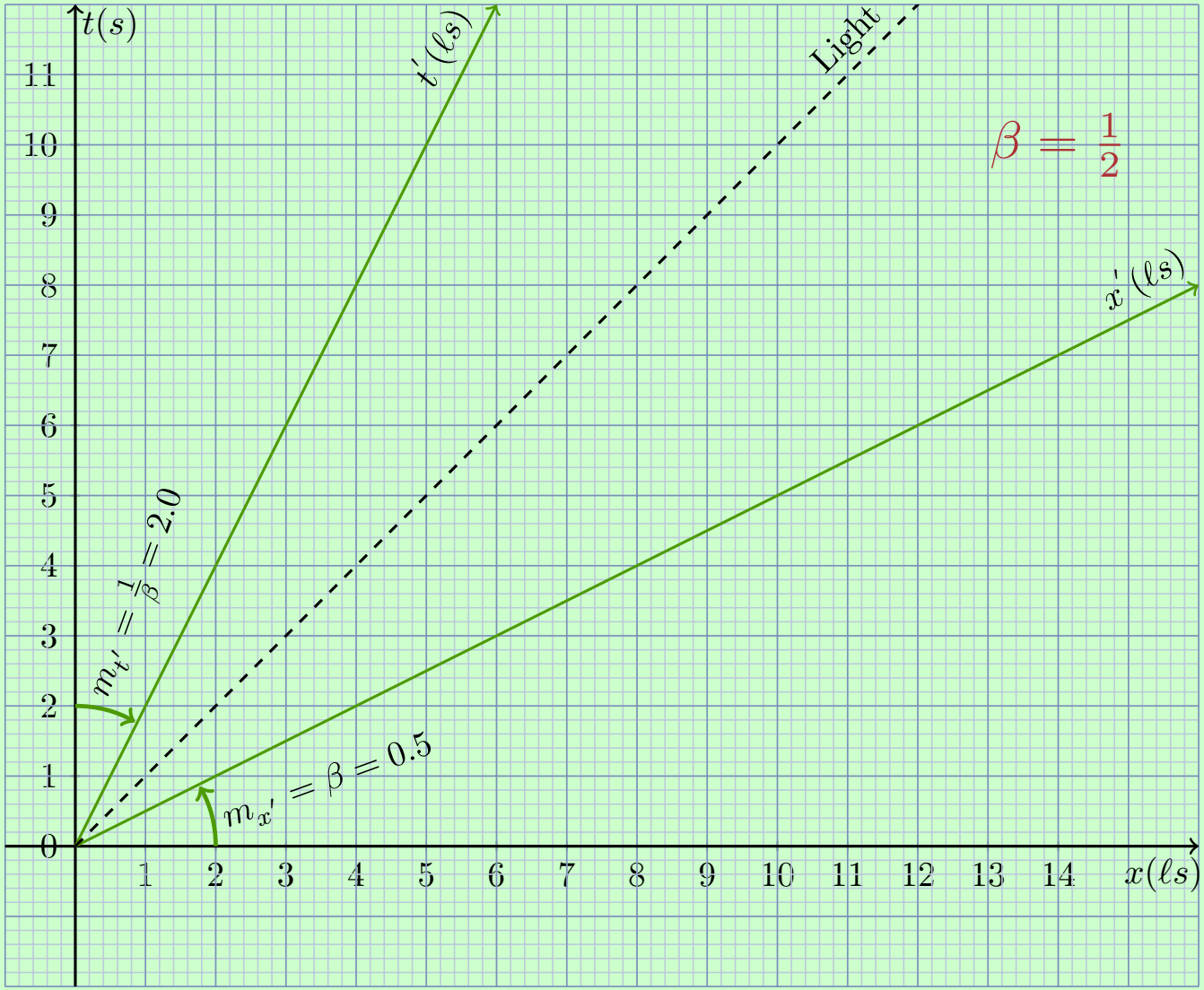
Spacetime Diagrams: Suppose I watch a ship zoom by at <u>β = 0.5</u>

t′ = Worldline
= location of ship
according to me

Slope of **t′ = 1/β**

x′ = "Nowslice"
= location of now
according to ship

Slope of **x′ = β**



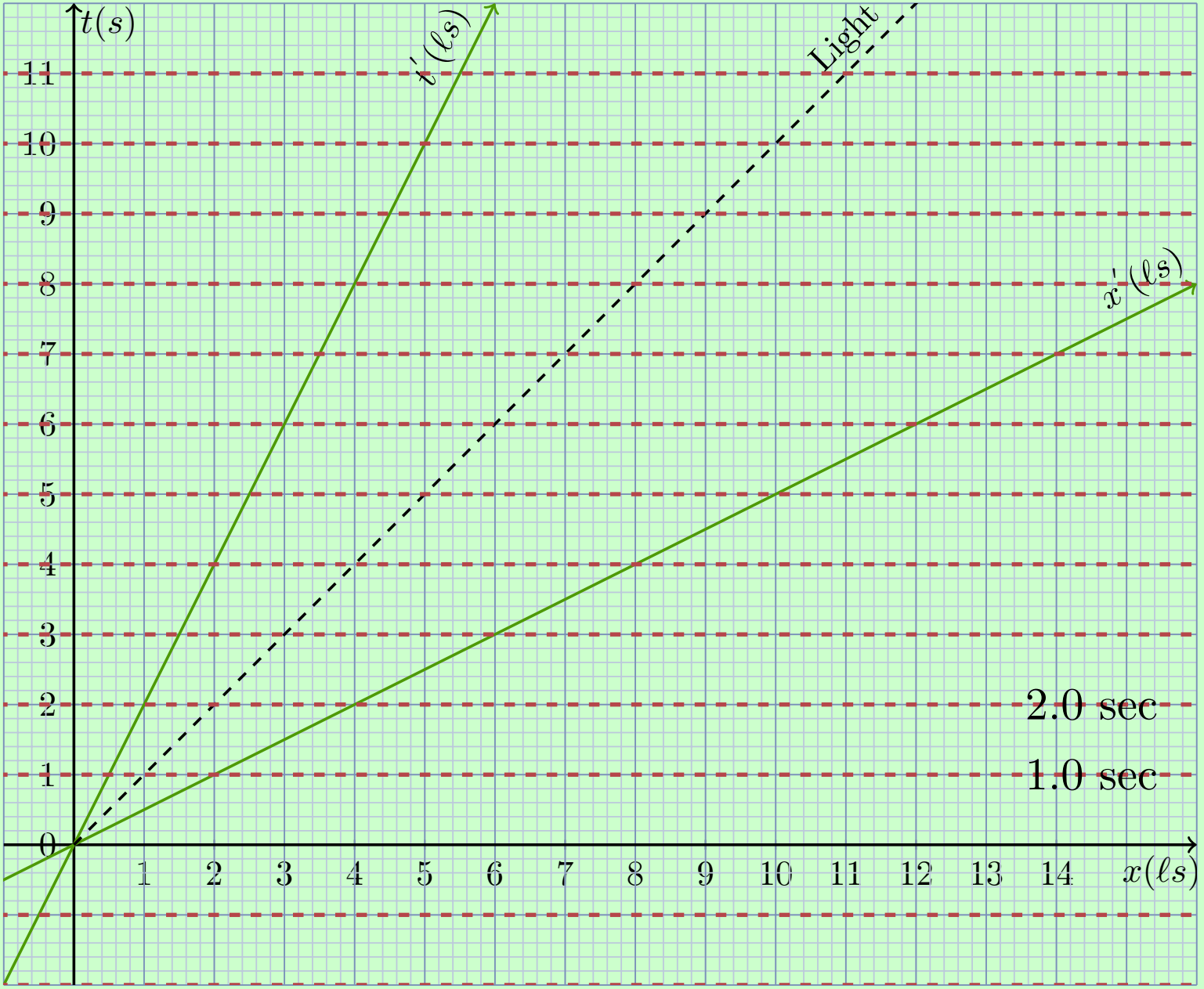
Ex:
Find the worldline and nowslice slopes

The t′ axis rotates in the opposite direction of x′ above

$$\Delta s^2 = \Delta x^2 - \Delta t^2$$ (That's why the minus)

My Now-Slices

Now-slices
Are parallel
to x-axis

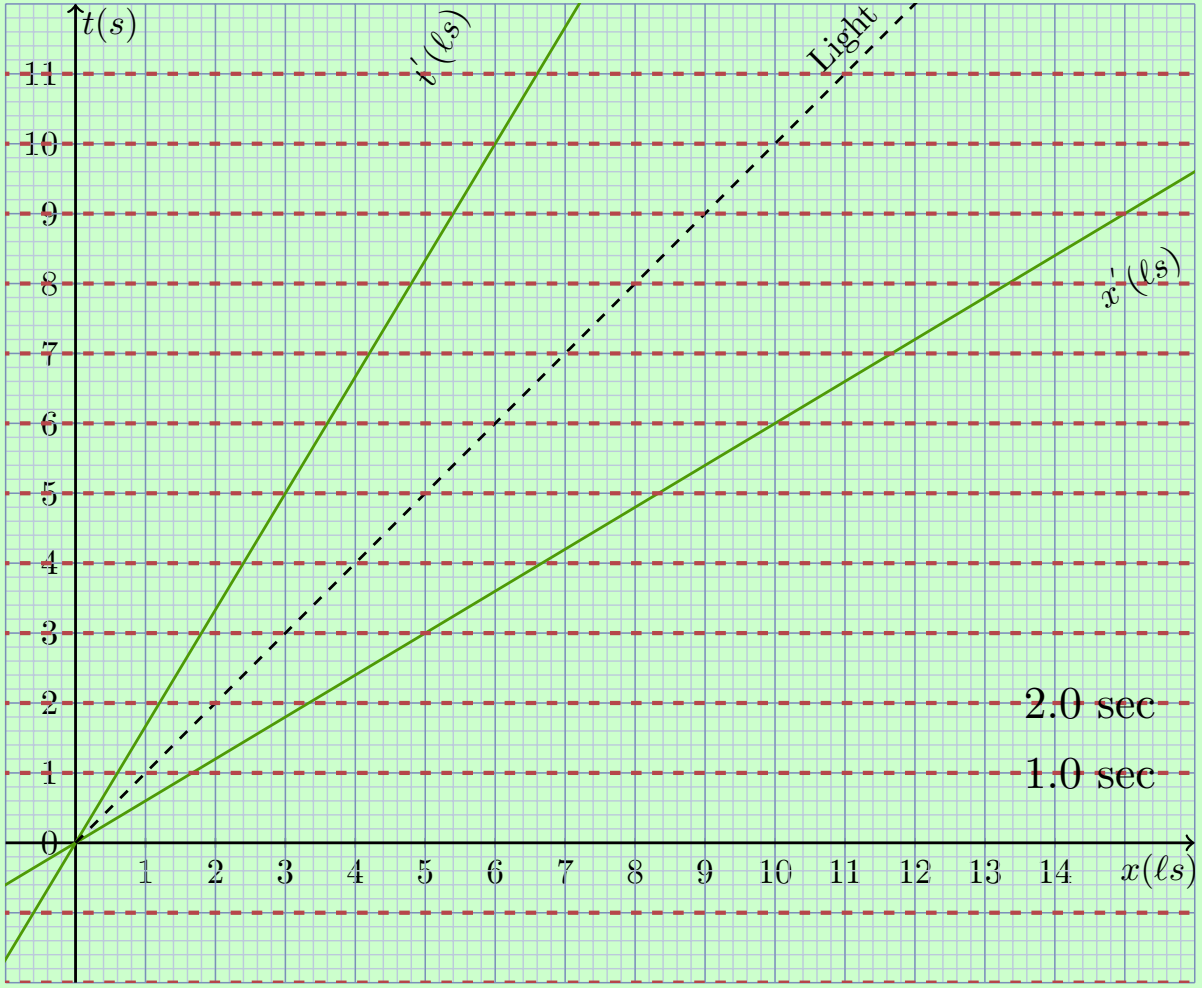


Quick Quiz

Draw a ST diagram for β = 3/5

ST diagram with
- Worldline
- My nowslices

What time is it on the ship?



Step 1. Calibrate t′ timeline: (Pick time t= 5, what time on ship?)

$$t = \gamma\tau$$

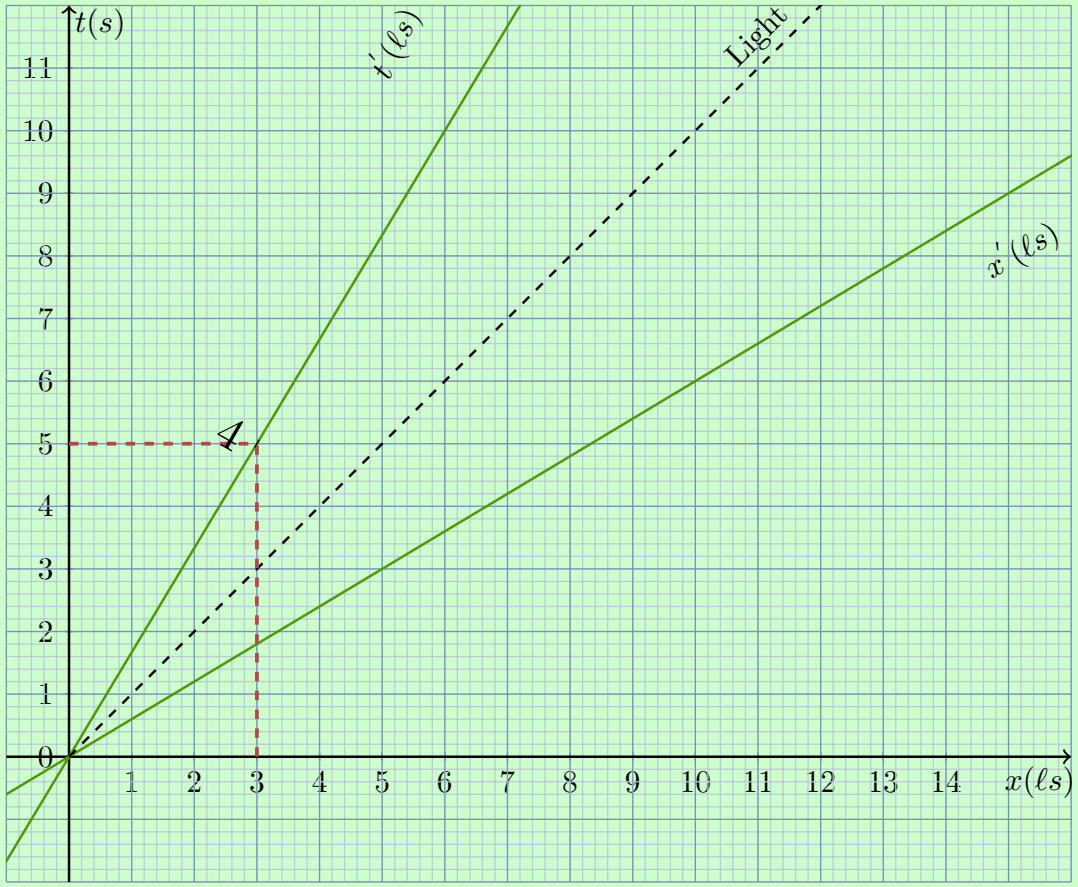
$$\gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{1-(3/5)^2}} = \frac{1}{\sqrt{1-9/25}} = \frac{1}{\sqrt{16/25}} = \frac{1}{4/5} = 5/4$$

Plugging this in:

$$t = \gamma\tau$$

$$5 = \left(\tfrac{5}{4}\right)\tau$$

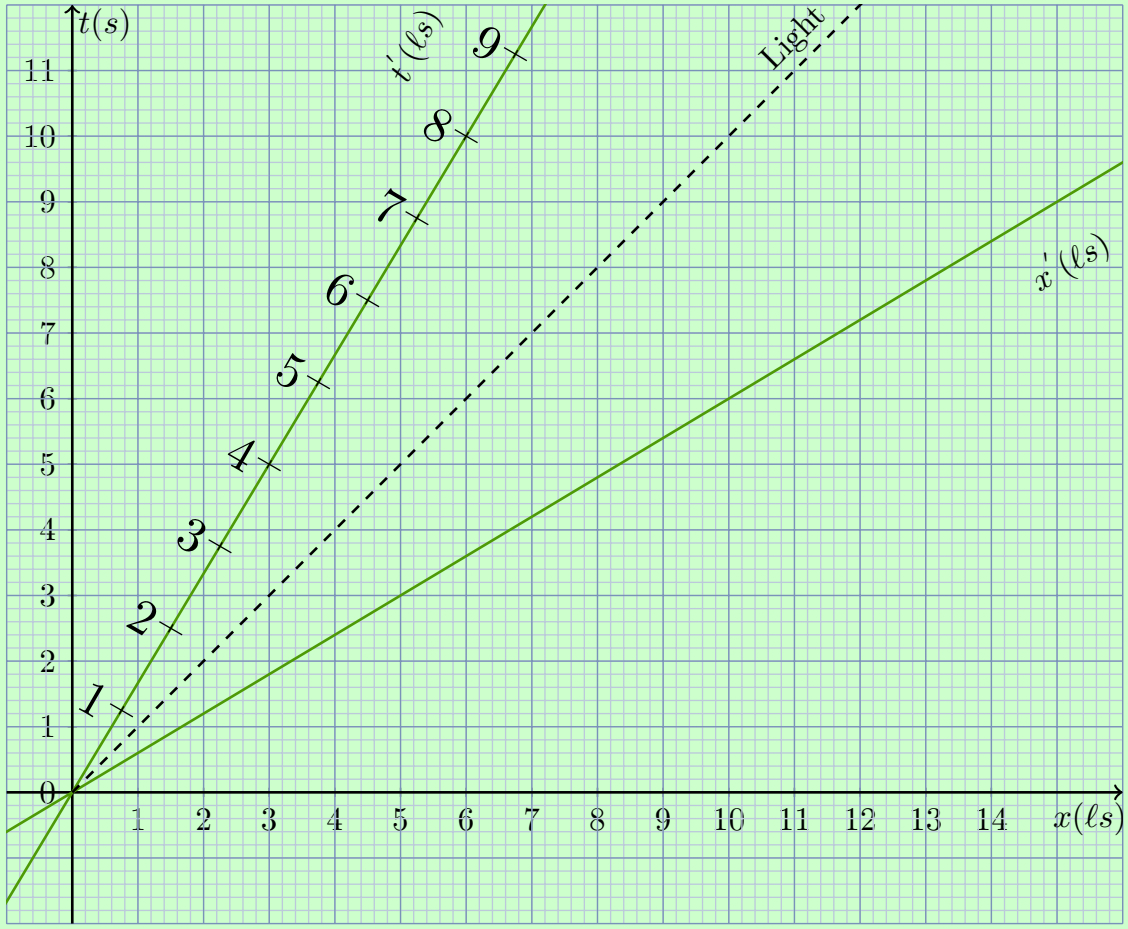
$$\boxed{\tau = 4}$$



Now use same spacing to finish whole axis….

(i.e. at 2.5 of my minutes Spaceship time will be 2.0 min)

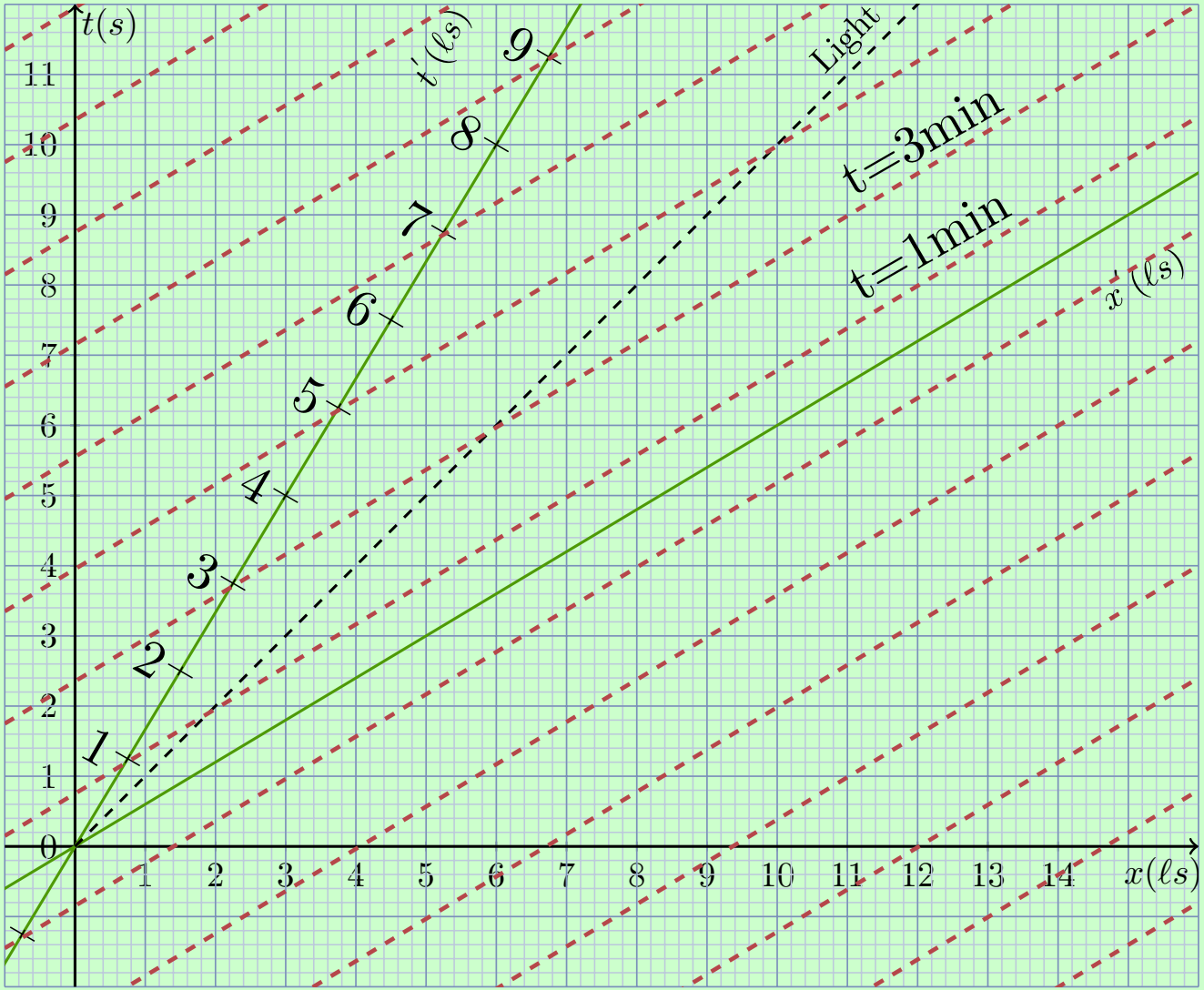
Now we know what time it is aboard the ship



What are the ship's now-slices?

In my frame now-slices parallel to x axis

In spaceship frame now-slices parallel to x′ axis



Now slice axis will have same distance between ticks